

Serial-Port Monitor in Visual Basic

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This software program can be used to monitor data being received or sent from or to a device connected to the COM port of your personal computer. RS-232 serial interface is also called COM port in short and it is one of the industry-standard device communication protocols. Developing hardware and software solutions for the COM port communication protocol, without using proper tools, is rather a challenging task. This program automatically detects the serial port in use and displays the communication details on screen with voice acknowledgement. This program allows developers to focus on their project instead of worrying about the hardware. It is similar to popular serial-communication programs such as HyperTerminal, Terminal and X-CTU, but with some unique features as explained below:

1. Full-duplex mode. With this, data can be transmitted in both directions at the same time

2. Flexible configuration. This software supports the baud rates up to 256,000, a number of data bits, a number of stop bits, different types to parity, flow control types and other settings. You can alter these settings as per your requirement

3. Save conversations to file. The software allows all the conversation, which is being transmitted and received, to be saved in a text file for future reference

4. Voice acknowledgement. The software is equipped with GUI and text-to-speech code for voice announcement, so that the user can enable the software to speak the message each time it receives or sends data

5. Speak and flash indication on each reception. The software has the option to flash the receive box along with the voice acknowledgement,

informing the user of the latest data being received

6. Send options. With this software, you can send arbitrary data on the serial port with resend option for the last sent data. It also has the option to send data multiple times in loops/repeatedly and at user-configurable time intervals

7. Time logging of every single conversation. The software allows to log the sent or received data along with the

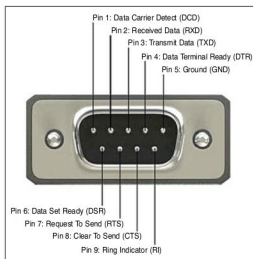


Fig. 1: D-type serial COM port connector

date and time

8. Available serial ports list. When the program is running, it checks the ports and generates a list of serial ports available on the computer

9. Remembers last configuration. The software remembers the last configurations and settings, and recovers them during the next start so that the user does not need to waste time in reconfiguring it

The serial port

The term 'serial' comes from the fact that a serial port 'serialises' data, i.e., it takes a byte of data and transmits the eight bits in the byte one at a time. The advantage is that a serial port needs only one wire to transmit the



eight bits whilst a parallel port needs eight wires. Before each byte of data is sent, a serial port sends a start bit, which is a single bit with a value of 0. After each byte of data, it sends a stop bit to signal that the byte is complete.

It may also send a parity bit. This is done to reduce and manage data errors during transmission.

Serial ports rely on a special controller chip, the universal asynchronous receiver/transmitter (UART), to function properly. The UART chip takes the parallel output of the computer's system bus and transforms it into the serial form for transmission through the serial port. In order to function faster, most UART chips have a built-in buffer of anywhere from 1 to 256kB or more. This buffer allows the chip to cache data coming in from the system

bus whilst it is processing data going out to the serial port. Whilst most standard serial ports have a maximum transfer rate of 115kbps (kilobits per second), high-speed serial ports, such as enhanced serial port (ESP) and super ESP, can reach data transfer rates of 460kbps.

The serial connection and data flow. Generally, the serial port has nine pins (or 25 pins) for external connections (refer Fig. 1). The primary use of a serial port is to connect a modem to a computer and the pin assignments reflect the same. Let us take a closer look at what happens at each pin when a modem is connected.

For a nine-pin connector, pins have the following designations: